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Engineering Reconstruction Analysis: Civilian Demilitarization and fabrication of the Fairchild Republic A-10 Airframe

Executive Summary

The Fairchild Republic A-10 Thunderbolt II serves as a singular case study in damage-tolerant aviation design. Conceived during the Cold War to survive high-intensity ground fire while delivering close air support (CAS), its architecture prioritizes structural redundancy, mechanical simplicity, and field maintainability over the aerodynamic refinement typical of high-speed interceptors. This report provides an exhaustive technical analysis for the theoretical reconstruction of an A-10 airframe utilizing civilian-authorized components, raw materials, and assembly methodologies. The scope of this document is strictly limited to the non-weaponized, airworthiness-critical aspects of the vehicle, focusing on the metallurgical composition, mechanical interconnects, and subsystem integration required to replicate the flight characteristics of this airframe in a civilian context.

Rebuilding the A-10 outside of a military supply chain presents unique engineering challenges, particularly regarding the substitution of restricted avionics and the management of mass properties following the removal of the GAU-8/A Avenger cannon system. The analysis indicates that approximately 80% of the airframe's structural weight comprises aluminum alloys, specifically the 2000 and 7000 series, while the remaining 20% consists of high-strength steel and titanium alloys critical for high-stress nodes and pilot protection. The propulsion system, centered on the General Electric TF34 turbofan, shares significant architectural commonality with the civilian CF34 series, providing a viable pathway for powerplant acquisition and sustainment.

This document is structured to serve as a comprehensive Bill of Materials (BOM) and fabrication guide. It dissects the aircraft into its elemental raw materials, analyzes the fastener technology required to join them, and specifies the finished mechanical components necessary for flight, all cross-referenced against civilian aerospace standards and supply chains.

1. Metallurgical Architecture and Raw Material Requirements

The foundational integrity of the A-10 airframe relies on a specific metallurgical "diet" designed to balance weight, static strength, and fatigue resistance. Unlike modern aircraft that utilize extensive carbon-fiber composites, the A-10 is primarily a metallic structure. This necessitates a precise understanding of the aluminum, titanium, and steel alloys required for fabrication, including their temper states and processing requirements.

1.1 Aluminum Alloys: The Structural Primary

The vast majority of the A-10's wetted surface and internal substructure is fabricated from aluminum. The selection of specific alloys is driven by the load profile of each component—tension-dominated structures require different metallurgical properties than compression-dominated ones.

1.1.1 Aluminum 2024-T3/T351 (Tension Members and Skins)

The lower wing skins and the majority of the fuselage monocoque are subjected to cyclic tension loads during flight. For these applications, Aluminum 2024 in the T3 or T351 temper is the mandatory specification.

- **Metallurgical Rationale:** The 2024 alloy derives its strength primarily from copper (3.8-4.9%) and magnesium (1.2-1.8%) alloying elements. In the T3 temper (solution heat-treated, cold worked, and naturally aged), it exhibits high fracture toughness and slow fatigue crack growth rates. This is critical for the A-10, which operates in turbulent low-altitude environments that induce significant airframe flexing. The "damage tolerance" design philosophy relies on the material's ability to sustain cracks without catastrophic failure between inspection intervals.[1]
- **Civilian Sourcing and Form Factors:**
  - **Sheet Stock:** Used for fuselage skins, empennage skins, and fairings. Thicknesses range from 0.032 inches for localized fairings to 0.063 inches for structural skins.
  - **Plate Stock:** The lower wing skin of the A-10 is integral to its structural life. Modern "HOG UP" life-extension programs mandated the use of "thick wing" skins machined from heavy plate to improve fatigue life.[2] A civilian reconstruction would require purchasing Alclad 2024-T351 plate (up to 1.0 inch thick) for machining these variable-thickness skins.
- **Raw Resource Requirement:**
  - **Total Estimated Mass:** 4,500 lbs of 2024-series aluminum.
  - **Scrap Rate Factor:** A buy-to-fly ratio of 3:1 is estimated for machined skins, necessitating the procurement of approximately 13,500 lbs of raw plate and sheet.

1.1.2 Aluminum 7075-T6/T7351 (Compression Members and Spars)

The upper wing skins, wing spars, and heavy fuselage bulkheads experience compression loads. Aluminum 7075 is selected for these applications due to its superior static strength characteristics.

- **Metallurgical Rationale:** Alloying with zinc (5.1-6.1%) produces the 7000-series alloys, which offer the highest strength of common aerospace aluminum. However, 7075-T6 is susceptible to Stress Corrosion Cracking (SCC) and exfoliation corrosion. For thick sections like the main wing spars (Front, Center, and Rear spars) [3], the T7351 temper is preferred. This over-aged temper sacrifices some tensile strength (approx. 10-15%) compared to T6 but drastically improves resistance to SCC, a critical factor for an aircraft designed to operate in humid, austere environments.

- **Application Specifics:**
    - **Wing Spars:** The A-10 utilizes a multi-spar wing design. These spars are machined from long, heavy extrusions or billets of 7075.
    - **Bulkheads:** The load-bearing frames, particularly those transferring landing gear and wing loads to the fuselage, are machined from 7075-T7351 plate.
- **Raw Resource Requirement:**
    - **Total Estimated Mass:** 6,800 lbs of 7075-series aluminum.
    - **Form Factors:** Large cross-section extrusions (up to 20 feet in length) and heavy plate (up to 4 inches thick).

1.1.3 Aluminum 6061-T6 (Secondary Structure and Fluid Lines)

For components requiring complex welding, forming, or lower structural criticality, Aluminum 6061-T6 is the standard.

- **Metallurgical Rationale:** This alloy contains magnesium and silicon, making it heat-treatable and highly corrosion-resistant. It is weldable, unlike the 2024 and 7075 series, making it ideal for fluid reservoirs, brackets, and interior avionics racks.
- **Application Specifics:** Hydraulic hard lines (tubing), environmental control system (ECS) ducting, and non-structural mounting brackets.

1.2 Titanium Alloys: The Structural Backbone and Armor

The A-10 is famous for its "Titanium Bathtub," but titanium's role extends beyond ballistic protection. It is used in high-stress fatigue zones and areas requiring high thermal resistance.

1.2.1 Titanium Ti-6Al-4V (Grade 5)

This alpha-beta alloy is the workhorse of the aerospace titanium market, accounting for the vast majority of titanium usage in the airframe.

- **The "Bathtub" (Cockpit Enclosure):** The pilot enclosure is a bolted assembly of titanium plates ranging from 0.5 inches to 1.5 inches in thickness.[2] While its primary military function is armor against 23mm projectiles [4], structurally, it serves as the forward fuselage's primary load-bearing node. It connects the nose landing gear and the GAU-8 gun mount to the center fuselage.
    - **Reconstruction Necessity:** In a civilian rebuild, this structure acts as a critical counterweight (ballast) to offset the removal of the gun and provides unmatchable firewall protection. Substituting it with aluminum would require a complete redesign of the forward fuselage rigidity and mass balancing.
- **Engine Nacelle Keels:** The rear fuselage supports the two TF34 engines. The keels and

firewalls in this section are fabricated from titanium to withstand the thermal soak from the engines and provide fire containment. Titanium's melting point (~1,660°C) is far superior to aluminum (~660°C).

- **Raw Resource Requirement:**
  - **Total Estimated Mass:** 1,500 lbs of Ti-6Al-4V.
  - **Processing:** Fabrication requires slow-speed machining with carbide tooling and waterjet cutting for plate stock. Hot forming is required for curved armor sections.

1.3 High-Strength Steels: Landing Gear and Fasteners

Where volume is limited and loads are extreme, high-strength steels are employed.

1.3.1 4340M and 300M Low-Alloy Steels

- **Landing Gear:** The A-10's landing gear struts must absorb high sink rates typical of short-field landings. These components are forged from 300M steel (a modified 4340 with silicon) and heat-treated to a tensile strength range of 280-300 ksi (kilopounds per square inch).[1]
- **Wing Attachment Pins:** The connection between the wing and the fuselage (at Wing Station 23) utilizes large-diameter steel pins.[1] These are critical fracture-critical components.
- **Processing:** These steels require Vacuum Arc Remelting (VAR) to ensure purity and are often cadmium or zinc-nickel plated to prevent corrosion. Baking after plating is mandatory to prevent hydrogen embrittlement.
- **Raw Resource Requirement:**
  - **Total Estimated Mass:** 2,200 lbs of steel forgings and bar stock.

1.3.2 Stainless Steel (17-4 PH)

Precipitation-hardening stainless steel is used for hydraulic fittings, control rod terminals, and engine mount components where corrosion resistance and high strength are required without plating.

1.4 Non-Metallic Materials

- **Composites:** While not a composite aircraft, the A-10 uses fiberglass/epoxy laminates for wingtips, the vertical stabilizer cap, and leading-edge slats where radar transparency or complex compound curves are required.[5]
- **Glazing:** The bubble canopy is a monolithic stretch-formed acrylic or polycarbonate unit, designed to offer a panoramic field of view.[4]
- **Fuel Tank Foam:** The internal fuel tanks are stuffed with reticulated polyurethane foam (Mil-Spec) to suppress explosions and reduce fuel sloshing.[4]

Table 1: Raw Material Bill of Materials (BOM) for Civilian

Reconstruction

Material Category	Alloy / Spec	Temper	Primary Application	Estimated Raw Weight (lbs)	Form Factor
Aluminum	2024	T3 / T351	Fuselage Skins, Lower Wing Skins	4,500	Alclad Sheet / Plate
Aluminum	7075	T6 / T7351	Wing Spars, Bulkheads, Ribs	6,800	Extrusions / Plate
Aluminum	6061	T6	Fluid Lines, Brackets, Ducts	800	Tube / Sheet
Titanium	Ti-6Al-4V	Annealed	Cockpit "Bathtub", Engine Keels	1,500	Plate / Billet
Steel	300M / 4340	Heat Treated	Landing Gear Struts, Axles	2,200	Forgings
Stainless Steel	17-4 PH	H1025	Hydraulic Fittings, Pins	600	Bar Stock
Polymer	Polycarbonate	Optical	Canopy Glazing	300	Cast Sheet
Elastomer	Buna-N / EPDM	Cured	Seals, Tires, Hoses	400	Molded / Extruded

Foam	Reticulated PU	Inert	Fuel Tank Baffling	200	Block Foam

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2. Structural Assembly: Fasteners, Junctions, and Bonding

The structural integrity of the A-10 is defined not just by the materials used, but by how they are joined. The airframe utilizes a mechanical fastening approach, typical of its era, relying on thousands of rivets, bolts, and specialized pins. Understanding the specific connector types is vital for a reconstruction project, as the fatigue life of the airframe is often determined by the quality of these junctions.

2.1 Fastener Technology and Standards

The reconstruction requires adherence to Military Standards (MS), Air Force-Navy (AN), and National Aerospace Standards (NAS). A single wing may contain tens of thousands of individual fasteners, each selected for specific shear and tension properties.

2.1.1 Solid Rivets: The Primary Skin Connector

The majority of the sheet metal assembly is joined using solid aluminum rivets. These require access to both sides of the structure (one side to apply the rivet gun, the other to hold the bucking bar).

- **Universal Head Rivets (MS20470):**
  - **Material:** 2117-T4 (AD) Aluminum. Identified by a small dimple in the center of the head.
  - **Application:** Internal structures, bulkheads, and areas where aerodynamic smoothness is not critical. The AD alloy is driven "cold" (without heat treatment) and work-hardens during installation.
- **Flush Head Rivets (MS20426):**
  - **Geometry:** 100° countersunk head.
  - **Application:** External skins (wings, fuselage) to maintain aerodynamic smoothness.
  - **Installation Method:** For thin skins (<0.040"), the skin is "dimpled" using dies. For thicker skins (>0.040"), the skin is machine countersunk. This distinction is critical to prevent knife-edge fatigue cracks in the skin holes.[6]

2.1.2 High-Strength Structural Fasteners

In areas where loads exceed the capability of aluminum rivets, or where high clamping force is required, specialized steel and titanium fasteners are employed.

- **Hi-Lok Fasteners (HL Series):** The A-10 makes extensive use of Hi-Lok fasteners, particularly the **HL844** series.[7]

- ○ **Mechanism:** A Hi-Lok consists of a threaded pin and a collar. The collar has a hexagonal wrenching element that shears off automatically when the correct torque is reached. This ensures consistent preload and prevents over-torquing.
- ○ **Application:** Spar-to-skin attachments, heavy bulkhead assembly, and wing root fittings.
- ○ **Civilian Sourcing:** HL844 pins (High-Strength Steel or Titanium) and matching collars (HL84/HL86) are available through aerospace distributors like Skybolt and Gen-Aircraft Hardware.[9]

- **Jo-Bolts and Blind Bolts:** In closed structures like the wing box or empennage where there is no access to the back side, high-strength blind bolts (NAS1669 or similar) are used. These expand on the blind side to form a head, clamping the sheets together.

2.1.3 Wing Attachment Pins (The Critical Junction)

The interface between the wing and the fuselage—specifically at Wing Station 23 (WS23)—is the most critical junction in the aircraft.[1]

- **Type:** Trunnion-to-clevis joint.
- **Hardware:** Large diameter (approx. 1.5 to 2.0 inch) precision-ground steel pins. These are often made of 17-4 PH stainless or plated 4340 steel.
- **Assembly:** The pins are installed with a close-tolerance fit (often requiring freezing the pin and heating the lug) to ensure zero play. They are secured with retaining nuts and cotter pins.

2.2 Junction Methodologies

2.2.1 Wing-to-Fuselage Integration

The A-10 utilizes a "continuous wing" concept where the center wing section passes through the lower fuselage.

- **Banjo Frames:** The fuselage frames at the wing intersection are shaped like banjos to allow the wing spars to pass through.
- **Load Path:** Flight loads are transferred from the wing skins to the spars, then through the banjo frames into the fuselage monocoque.
- **Sealant:** All faying (mating) surfaces in this junction must be sealed with a corrosion-inhibiting sealant (Mil-PRF-81733) to prevent fretting corrosion caused by micro-movements during high-G maneuvers.

2.2.2 The Titanium Armor Assembly

The "bathtub" is not a welded unit; welding titanium armor degrades its ballistic properties.

- **Bolted Assembly:** The plates are joined using heavy titanium angles and splice plates.
- **Fasteners:** Titanium or A286 stainless steel bolts are mandatory to prevent galvanic corrosion. Cadmium-plated steel bolts would react with the titanium, leading to rapid

embrittlement.

2.3 Chemical Bonding and Sealants

- **Fuel Tank Sealing:** The A-10 features integral fuel tanks ("wet wings"). The structure itself forms the tank walls.
  - **Material: Polysulfide Sealant (Class B)**. The industry standard is **BMS 5-45** or **PR-1440**.[10] This sealant is applied to all structural joints, fasteners, and seams within the fuel boundary to prevent leaks.
  - **Application:** Fillet sealing is performed over all fastener tails inside the tank.
- **Faying Surface Sealing:** Sealing mating surfaces protects against moisture intrusion. This is vital for the 7075-T6 components which are prone to stress corrosion cracking.

Table 2: Fastener and Connector Bill of Materials

Component Type	Standard / Series	Material	Application	Notes
Solid Rivet	MS20470AD	2117-T4 Aluminum	Internal Structure	Universal Head
Solid Rivet	MS20426AD	2117-T4 Aluminum	External Skins	Flush (Countersunk)
Pin-Rivet	HL844 / HL10 [7]	Alloy Steel / Ti	High-Stress Joints	Constant Torque (Hi-Lok)
Blind Rivet	NAS1919 / CherryMAX	Monel / Steel	Close-out Panels	Structural Blind Rivet
Wing Pin	Custom / NAS	4340 Steel	Wing-Fuselage Joint	Precision Ground
Hydraulic Fitting	AN815 / AN818 [12]	Steel / Aluminum	Fluid Lines	37° Flare JIC Standard
Sealant	BMS 5-45 / PR-1440	Polysulfide	Fuel Tanks	Fuel Resistant [11]

Connector	MIL-DTL-38999	Composite / Metal	Avionics Wiring	Circular Connector [13]

3. Propulsion Systems: The TF34/CF34 Nexus

The propulsion system is the heart of the A-10's capability. The aircraft is powered by two General Electric TF34-GE-100 high-bypass turbofans. For a civilian reconstruction, the relationship between this military engine and its civilian counterpart, the CF34, provides a viable logistical path.

3.1 General Electric TF34-GE-100

- **Specifications:**
  - **Thrust:** ~9,065 lbs (40.3 kN) each.[14]
  - **Bypass Ratio:** ~6:1. This high bypass ratio is unusual for combat aircraft but provides the A-10 with its exceptional loiter time and fuel efficiency.
  - **Configuration:** Single-stage fan, 14-stage high-pressure compressor, annular combustor, 4-stage low-pressure turbine.
- **Civilian Cross-Reference:** The TF34 is the parent design of the **GE CF34** series, widely used on regional jets like the Bombardier Challenger 601/604 and the CRJ100/200 series.[15]
  - **CF34-3:** This engine shares the same core (compressor, combustor, HPT) as the TF34.
  - **Reconstruction Strategy:** A civilian builder can source timed-out or serviceable CF34-3 engines. While the accessory gearbox location and mounting points differ slightly, the core performance and envelope are nearly identical. Using a CF34-3 would likely be easier for certification (civilian Type Certificate exists) than trying to demilitarize a TF34.

3.2 Nacelle and Mounting Structure

The A-10's engines are pod-mounted on pylons above the rear fuselage. This configuration shields the engines from Foreign Object Damage (FOD) during operation from rough strips and masks the infrared signature from ground-based threats.

- **Nacelle Construction:**
  - **Inlet Cowl:** Aluminum skin with acoustic treatment (honeycomb core) to reduce noise.
  - **Firewalls:** Titanium barriers separating the engine from the airframe.
  - **Access Panels:** Large, hinged cowling doors allowing complete access to the engine core for maintenance.
- **Mounting Pylons:** The pylons are structural cantilevers attached to the rear fuselage frames. They must withstand the engine's thrust, weight, and gyroscopic loads during maneuvering.
  - **Vibration Isolation:** The engine attaches to the pylon via Lord Mounts (vibration

isolators) [9] to prevent engine harmonics from damaging the airframe.

3.3 Fuel System

- **Capacity:** Approximately 11,000 lbs (1,600 gallons) of internal fuel split between two fuselage tanks and two wing tanks.
- **Tank Construction:** The A-10 uses self-sealing bladder cells.
- **Inerting:** The tanks are filled with reticulated polyurethane foam.[4] This foam prevents fuel sloshing and acts as a 3D flame arrestor, preventing catastrophic explosion if a tank is penetrated.
- **Pumps:** The system relies on boost pumps to feed the engines.
  - **Part Number: Argo-Tech 724400-2** [17] is the OEM boost pump.
  - **Civilian Equivalent:** Similar submerged centrifugal boost pumps are used on the Boeing 737 and can be adapted with appropriate flange modifications.[18]

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4. Hydromechanical Systems: Power and Control

The A-10 is a hydraulically muscled machine. Its flight controls, landing gear, brakes, and gun drive (in the military version) are all hydraulically actuated.

4.1 Hydraulic Power Generation

The aircraft features two independent, redundant hydraulic systems (Left/System A and Right/System B) operating at **3,000 PSI**.

- **Primary Pumps:** Each engine drives a variable-displacement axial piston pump.
  - **Part Number: Vickers PVB-044-2**.[19]
  - **Specifications:** Rated for 3,000 PSI, utilizing MIL-H-5606 fluid.
  - **Civilian Availability:** This specific pump series is widely used in industrial and commercial aviation applications and is readily rebuildable.
- **Fluid: MIL-H-5606** (Red Oil). This is a mineral-based fluid. While flammable, it offers excellent lubricity and low-temperature viscosity.
  - *Civilian Note:* Conversion to fire-resistant Skydrol (purple) is theoretically possible but would require changing every seal in the aircraft to EPDM rubber, which is chemically incompatible with 5606 seals (Buna-N). It is recommended to stay with 5606 for compatibility with vintage components.

4.2 Flight Controls and Manual Reversion

The primary flight controls (ailerons, elevators, rudders) are hydraulically boosted. However, the A-10 is unique in its "Manual Reversion" capability.[4]

- **Hydraulic Mode:** Pilot input opens servo valves, directing fluid to actuators that move the surfaces.

- **Manual Reversion Mode:** If hydraulic pressure is lost, the pilot mechanically flies the aircraft.
  - **Mechanism:** The control stick is linked via cables to "servo tabs" (small control surfaces on the trailing edge of the main surfaces). Moving the stick moves the tab, which creates aerodynamic force to move the main surface "backwards" relative to the tab.
  - **Hardware:**
    - **Cables:** 7x19 Stainless Steel Aircraft Cable (MIL-DTL-83420).
    - **Pulleys:** Phenolic or aluminum pulleys (MS20219 / MS20220).
    - **Turnbuckles:** Brass/Steel turnbuckles (MS21251) for tensioning the cables.

4.3 Landing Gear

The A-10 uses a robust tricycle gear designed for rough fields.

- **Main Landing Gear (MLG):** Retracts forward into wing pods. The wheels remain partially exposed to act as skid plates in a belly landing.
  - **Actuator: Teijin Seiki 1523100-4** [21] hydraulic actuator.
  - **Tires: 36 x 11 inch** Type VII tires.[22] These are high-flotation tires available from Goodyear or Michelin.
- **Nose Landing Gear (NLG):** Retracts fully into the fuselage. Offset to the starboard side to clear the gun.
  - **Tires: 24 x 7.7 inch** tires.[22]
- **Brakes:** Multi-disc hydraulic brakes with an anti-skid system. Carbon brake upgrades (common on modern CF34-powered jets) could be adapted to save weight.

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5. Avionics, Electrical, and Environmental Systems

Demilitarizing the A-10 requires replacing its combat-centric avionics with a civilian-compliant suite capable of IFR (Instrument Flight Rules) operation.

5.1 Electrical Power

- **Generation:** Two 30/40 kVA Integrated Drive Generators (IDGs) providing **115V AC / 400Hz** 3-phase power.
- **DC Power:** Transformer-Rectifier Units (TRUs) convert AC to **28V DC** for cockpit instrumentation and avionics.
- **Wiring:** The original Kapton wiring used in A-10s is prone to aging and arc faults. A comprehensive rebuild must utilize **Tefzel (M22759/34)** wire [23], which is the modern FAA standard for durability and safety.

5.2 Avionics Retrofit (Glass Cockpit)

The original A-10A utilized steam gauges, and the A-10C uses proprietary military computers (CICU) that are unobtainable. A civilian rebuild should leverage the "Experimental Exhibition" category to install modern commercial avionics.

- **Primary Flight Display (PFD): Garmin G600 TXi**. This large touchscreen unit integrates attitude, airspeed, altitude, and moving maps. It supports the high-vibration environment of the A-10.
- **Navigation: Garmin GTN 750Xi**. A GPS/NAV/COM unit that provides IFR navigation, communication, and transponder control.
- **Engine Monitoring:** A custom DAU (Data Acquisition Unit) would be needed to convert the analog signals from the TF34 engine sensors (N1, N2, ITT, Oil Pressure) into digital ARINC 429 data for display on the Garmin system.
- **Connectors:** All avionics wiring should use **MIL-DTL-38999 Series III** circular connectors.[13] These are the industry standard for high-reliability connections in harsh environments.

5.3 Environmental Control System (ECS)

The pilot needs pressurization and temperature control, provided by bleed air from the engines.

- **Air Cycle Machine (ACM):** This turbomachine cools the hot bleed air.
  - **Civilian Equivalent:** Units manufactured by **Fairchild Controls** (now Collins Aerospace) or Honeywell for business jets (like the Citation VII) are sized similarly.[24]
- **Pressure Regulation:** Pneumatic pressure regulators (e.g., **Fairchild Model 10** series) [25] are used to step down the high-pressure engine bleed air to usable levels for the cockpit and fuel tank pressurization.

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6. Critical Integration Challenges and Feasibility

Rebuilding the A-10 is an exercise in systems integration. Two major hurdles exist:

6.1 The "Gun Problem": Mass Properties and CG

The **GAU-8/A Avenger** cannon system (gun, drum, and feed) weighs approximately 4,000 lbs and is located in the extreme nose.

- **The Issue:** Removing the gun shifts the Center of Gravity (CG) dramatically aft. The A-10 would become tail-heavy and uncontrollable (unstable in pitch).
- **The Solution:** Ballast. The civilian A-10 must carry a "dummy gun"—a mass simulator made of steel or lead installed in the nose to replicate the gun's mass and moment.
- **Alternative:** Relocating heavy systems (batteries, hydraulic reservoirs) to the nose can offset some weight, but dead ballast will likely still be required.

6.2 Structural Life and Fatigue

The A-10 fleet has historically suffered from wing fatigue cracking.

- **Rebuild Requirement:** Any civilian A-10 project must utilize the **"Thick Wing"** structural standard. This involves using thicker skins on the lower wing panels and cold-working all fastener holes to induce compressive stresses that retard crack growth.
- **Inspection:** A baseline Non-Destructive Testing (NDT) survey using Eddy Current and Ultrasound is mandatory for all wing spar caps and the WS23 attachment pins before assembly.

7. Comprehensive Finished Parts List (Civilian Authorized)

System	Component Name	Manufacturer / OEM	Part Number / Standard	Civilian Source / Equivalent
Propulsion	Turbofan Engine	General Electric	TF34-GE-100 / CF34-3	Bombardier Challenger 604 MRO
Hydraulics	Main Pump	Vickers (Eaton)	PVB-044-2 [19]	Industrial Hydraulics Suppliers
Hydraulics	Landing Gear Actuator	Teijin Seiki	1523100-4 [21]	Aircraft Salvage / MRO
Hydraulics	Pressure Valve	Benbow Mfg	11300 (204-076-343-3)	BAS Part Sales [19]
Hydraulics	Servo Cylinder	Conair Inc.	1660-17 [19]	Surplus / Rebuild
Landing Gear	Main Tire	Goodyear	36x11 Type VII	Commercial Aviation Tire Dist.

Landing Gear	Nose Tire	Goodyear	24x7.7 Type VII	Commercial Aviation Tire Dist.
Fuel	Boost Pump	Argo-Tech	724400-2 [17]	Boeing 737 Fuel System Parts
Fuel	Sealant	PPG Aerospace	BMS 5-45 / PR-1440	Chemical Suppliers [11]
ECS	Pressure Regulator	Fairchild Controls	Model 10 Series [25]	Pneumatic Control Suppliers
Fasteners	Hi-Lok Pin	Hi-Shear	HL844-10-13 [7]	Skybolt / Gen-Aircraft Hardware
Fasteners	Structural Bolt	NAS	NAS6203 / NAS6604	Aerospace Hardware
Fasteners	Flush Rivet	MS Standard	MS20426AD	Aircraft Spruce / Wicks
Avionics	Connector	Amphenol / Souriau	MIL-DTL-38999 Ser. III	Electronics Distributors (DigiKey/Mouser)

8. Conclusion

The reconstruction of a Fairchild Republic A-10 Thunderbolt II using civilian-authorized components is a formidable engineering endeavor, yet physically feasible. The aircraft's reliance on aluminum alloys (2024/7075) and mechanical fasteners (MS rivets/Hi-Loks) makes it more accessible to fabrication than modern composite airframes requiring autoclaves. The propulsion challenge is solved by the widespread availability of the GE CF34 engine family, which shares the TF34's DNA.

However, the demilitarization process introduces significant complexity in mass balancing. The "Titanium Bathtub" must be retained not just for heritage, but as a critical structural node and counterweight. The avionics must be completely reimagined, replacing obsolete military computers with modern glass cockpit solutions like the Garmin G600. Ultimately, a civilian A-10 represents the pinnacle of "warbird" restoration—a machine built to tolerate the worst of the battlefield, repurposed to demonstrate the enduring principles of survivability and mechanical resilience.

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